E-COMMERCE SALES ANALYTICS

Funnel Analysis & Customer Behaviour

Project Report

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Project Domain: E-commerce | Data Analytics | Business Intelligence

Tools & Technologies: PostgreSQL, SQL, Microsoft Excel, Power BI

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1. Executive Summary

The E-commerce Sales Analytics project focuses on understanding customer purchasing behaviour, evaluating sales performance, and identifying opportunities to improve business conversion rates and customer engagement.The project involves analyzing e-commerce transactional and customer interaction data to understand how users move through different stages of the purchasing journey, which acquisition channels generate the highest revenue, and how customer behaviour contributes to overall business performance. Using SQL and PostgreSQL, the data was explored, cleaned, validated, and analyzed to identify meaningful patterns across customer acquisition, sales performance, conversion funnels, and customer retention. Microsoft Power BI was subsequently used to transform analytical findings into interactive dashboards for business reporting and decision-making.

Key Findings

- **Customer Conversion:** The overall product-view-to-purchase conversion rate was 33.71%.
- **Funnel Performance:** The highest stage-wise conversion rate was observed between checkout initiation and purchase completion, at 73.45%.
- **Acquisition Performance:** Organic Search generated the highest revenue of approximately ₹23,205.55.
- **Customer Acquisition:** Organic Search also recorded the highest number of customers, with 295 customers.
- **Customer Behaviour:** The analysis identified 326 retained customers and 64 churned customers within the project's retention analysis.

Business Value

The analysis provides insights into customer purchasing patterns, acquisition channel effectiveness, revenue contribution, and potential conversion improvement opportunities.

These findings can help e-commerce businesses make informed decisions regarding marketing investments, customer experience optimization, sales strategies, and customer retention initiatives.

The final deliverable consists of an analytical project report supported by interactive Power BI dashboards and SQL-based analysis.

2. Project Overview

The rapid growth of e-commerce has made data-driven decision-making essential for businesses operating in the digital marketplace. Every customer interaction, from browsing a product to completing a purchase, generates valuable information that can help businesses understand customer preferences and purchasing behaviour.

However, collecting data alone does not provide business value. Organizations need to analyze this information to identify trends, evaluate performance, understand customer drop-off points, and discover opportunities for improvement.

This project focuses on analyzing an e-commerce business dataset to evaluate sales performance, customer acquisition effectiveness, purchasing patterns, and the customer conversion journey.

The analysis follows a structured approach, beginning with data exploration, cleaning, and validation using SQL and PostgreSQL. Further analysis is performed to evaluate important business metrics, including revenue contribution, customer acquisition, conversion rates, and customer behaviour.

Finally, Power BI is used to develop interactive dashboards that present the analytical findings in an understandable and business-oriented format.

The overall objective is to transform raw e-commerce data into meaningful insights that can support strategic and operational business decisions.

3. Business Problem Statement

An e-commerce business generates large amounts of data through customer interactions, marketing campaigns, product browsing activities, and completed transactions.

Despite having access to this information, businesses may face challenges in understanding why visitors do not complete purchases, which marketing channels generate the most valuable customers, and what factors influence overall sales performance.

Without proper analysis, businesses may struggle to:

- Identify customer drop-off points throughout the purchasing journey.
- Understand which acquisition channels generate the highest revenue.
- Evaluate the effectiveness of different customer acquisition channels.
- Identify differences in customer purchasing behaviour.
- Monitor customer retention and repeat purchasing patterns.
- Discover opportunities to improve conversion rates and sales performance.

Problem Statement

The objective of this project is to analyze e-commerce customer interaction and sales data to identify patterns in customer purchasing behaviour, evaluate the effectiveness of acquisition channels, measure conversion funnel performance, and assess customer retention.

The analysis aims to uncover actionable business insights that can help improve customer conversion, optimize marketing performance, increase revenue, and support better business decision-making.

Key Business Questions

The project is designed to answer the following questions:

1. How effectively are visitors progressing through the e-commerce purchasing funnel?
2. At which stage of the purchasing journey are customers most likely to drop off?
3. Which customer acquisition channel generates the highest revenue?
4. Which acquisition channel attracts the highest number of customers?
5. How does revenue performance vary across different acquisition channels?
6. Which devices are most commonly used by customers to browse products?
7. What proportion of customers are repeat customers versus one-time customers?
8. What patterns can be identified in customer retention and purchasing behaviour?
9. What business opportunities can be identified to improve overall e-commerce performance?

Expected Business Outcomes

- Better understanding of customer purchasing behaviour.
- Identification of conversion funnel inefficiencies.
- Improved visibility into acquisition channel performance.
- Identification of revenue-generating channels.
- Better understanding of customer retention patterns.
- Data-driven recommendations for improving sales and customer experience.

4. Project Objectives

- **Analyze customer purchasing behaviour.**
- **Evaluate conversion funnel performance.**
- **Measure sales and revenue performance.**
- **Compare customer acquisition channels.**
- **Identify high-performing products and categories.**
- **Understand customer retention and repeat purchasing.**
- **Develop interactive Power BI dashboards.**
- **Provide data-driven business recommendations.**

5. Dataset Description

The project uses five interconnected datasets maintained in separate Excel sheets.

Dataset	Purpose
Customers	Customer information and acquisition details
Events	Customer interactions and shopping activities
Orders	Transaction and order-level information
Order_Items	Individual products included in orders
Products	Product details and category information

These datasets were connected using relevant identifiers to support customer-level, product-level, and transaction-level analysis.

6. Tools and Technologies

Tool	**Application**
Microsoft Excel	**Initial data cleaning and preparation**
PostgreSQL	**Relational database management and validation**
SQL	**Data exploration, transformation, and analysis**
Power BI	**Data modeling, DAX measures, and visualization**

Project Workflow:
Excel → PostgreSQL → SQL Analysis → Power BI

7. Data Cleaning and Preparation

Data preparation was performed using Microsoft Excel and PostgreSQL.

Excel:

- Inspected dataset structures and column formats.
- Checked and handled missing values and duplicate records.
- Reviewed data consistency and corrected identified issues.
- Prepared datasets for database import.

PostgreSQL:

- Imported and validated all five datasets.
- Reviewed table structures and data types.
- Validated identifiers and table relationships.
- Checked record counts and data consistency.

8. Analytical Methodology

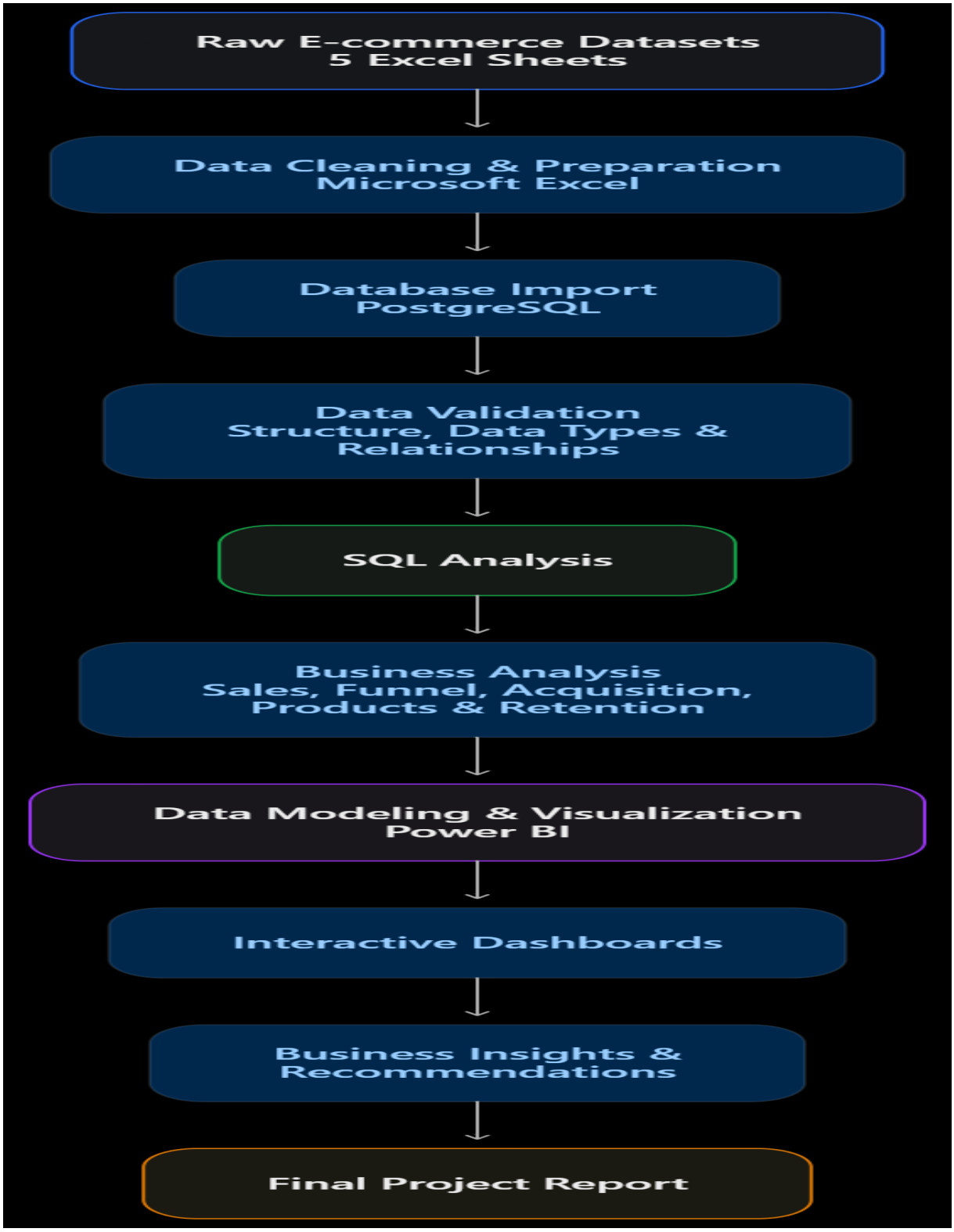
SQL was used to analyze the five datasets through joins, aggregations, CTEs, conditional logic, and window functions.

The analysis covered:

1. Sales and revenue performance.
2. Customer acquisition effectiveness.
3. Product and category performance.
4. Customer conversion funnel.
5. Customer retention and repeat purchasing.

The analytical results were subsequently used to develop interactive Power BI reports.

Project Diagram:



9. Key Performance Indicators (KPIs)

The following KPIs were used to evaluate business performance.

KPI	Business Purpose
Total Revenue	Measures overall sales performance
Total Orders	Measures transaction volume
Average Order Value (AOV)	Evaluates average revenue per order
Customer Conversion Rate	Measures purchasing effectiveness
Funnel Conversion Rate	Evaluates progression between funnel stages
Customer Retention	Evaluates repeat customer behaviour
Channel Revenue	Measures acquisition channel contribution

10. SQL Analysis and Key Findings

10.1 Conversion Funnel Analysis

The purchasing funnel was analyzed using four major customer interaction stages.

Funnel Stage	Unique Customers
Product Views	1,157
Add to Cart	730
Begin Checkout	531
Purchases	390

Conversion Performance:

Conversion Stage	Conversion Rate
View to Cart	63.09%
Cart to Checkout	72.74%
Checkout to Purchase	73.45%
Overall Conversion	33.71%

Key Insight: Approximately 66.29% of product viewers did not reach the purchase stage. The largest proportional conversion loss occurred between product viewing and cart addition.

Business Implication: Improving product discovery, product information, pricing transparency, and add-to-cart engagement could create opportunities to improve overall conversion.

10.2 Customer Acquisition Analysis

Acquisition channels were compared based on customer volume, purchasing performance, revenue, and average order value.

Channel	Customers	Purchasers	Conversion	Revenue
Organic Search	295	104	35.25%	₹23,205.55
Paid Search	243	73	30.04%	₹12,726.09
Direct	190	65	34.21%	₹12,526.22
Social Media	199	68	34.17%	₹12,145.85
Email	179	50	27.93%	₹9,302.13
Referral	94	30	31.91%	₹5,869.62

Key Insights:

- Organic Search generated the highest revenue and customer volume.
- Organic Search also recorded the highest conversion rate.
- Email recorded the lowest conversion rate among the analyzed channels.
- Direct traffic demonstrated competitive conversion performance.

Business Implication: Organic Search represents an important revenue opportunity. Email and Paid Search campaigns could benefit from further audience segmentation and campaign optimization.

10.3 Product and Category Performance

Product and category-level performance was evaluated using order and order-item information.

Key Finding: Electronics emerged as the highest revenue-generating product category in the analysis.

Business Implication: The business can further investigate product demand, pricing, inventory availability, and category-level profitability to optimize its product strategy.

10.4 Monthly Revenue Analysis

Monthly completed revenue was evaluated to understand sales trends.

Month (2025)	Revenue
January	₹663.41
February	₹4,430.30
March	₹5,558.15
April	₹9,178.61
May	₹18,342.58
June	₹37,602.41

Key Insight: Revenue demonstrated a strong upward trend during the analyzed six-month period, with June recording the highest revenue.

Business Implication: The business should investigate the factors behind this growth, including product demand, customer acquisition, promotions, and seasonal purchasing patterns.

10.5 Customer Retention Analysis

The customer purchasing analysis identified the following results:

Metric	Customers
Total Purchasing Customers	390
Retained Customers	326
Churned Customers	64
Retention Rate	83.59%
Churn Rate	16.41%

Key Insight: The analysis classified 83.59% of purchasing customers as retained, while 16.41% were classified as churned.

Business Implication: Customer engagement, repeat purchase incentives, and personalized marketing strategies could help strengthen customer retention.

11. Power BI Dashboard and Visualization

Power BI was used to transform the analytical results into interactive dashboards for business reporting.

The dashboards present key performance indicators, customer behaviour, acquisition performance, revenue contribution, and conversion funnel metrics.

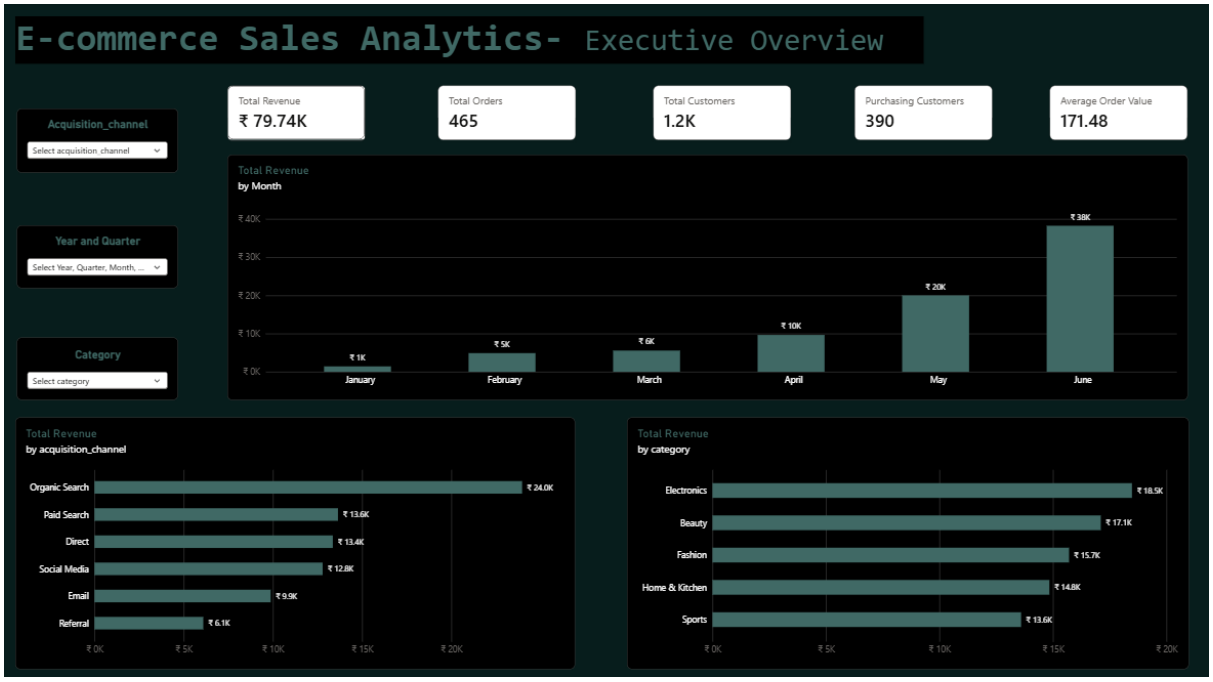


Figure 1: E-commerce Sales Analytics – Funnel and Customer Behaviour

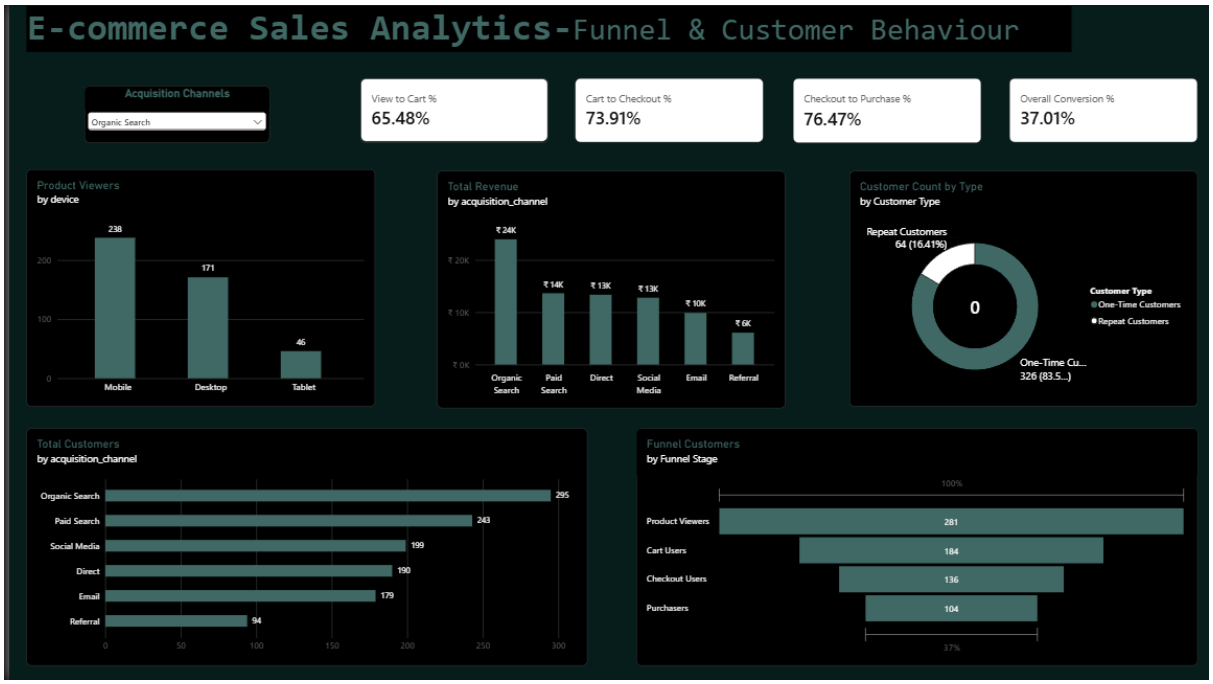


Figure 2: E-commerce Sales Analytics – Sales and Revenue Performance

12. Key Business Recommendations

Based on the analytical findings, the following recommendations are proposed:

1. Improve Funnel Conversion: Investigate product-page engagement and add-to-cart barriers to reduce early-stage customer drop-off.
2. Optimize Acquisition Channels: Continue strengthening Organic Search while evaluating the effectiveness and cost efficiency of other channels.
3. Improve Email Marketing: Review campaign targeting, content, and audience segmentation to improve email conversion performance.
4. Strengthen Product Strategy: Use category-level revenue insights to optimize product availability and marketing priorities.
5. Encourage Repeat Purchases: Introduce personalized offers, loyalty initiatives, and customer re-engagement campaigns.
6. Monitor Revenue Trends: Track monthly revenue and investigate the factors contributing to significant changes in sales performance.

13. Conclusion

The E-commerce Sales Analytics project demonstrates how SQL-based analysis and business intelligence tools can transform raw e-commerce data into meaningful business insights.

The analysis identified important patterns in customer conversion, acquisition channel effectiveness, revenue performance, product categories, and customer retention.

Organic Search emerged as the leading revenue-generating channel, while the conversion funnel highlighted opportunities to improve customer progression from product discovery to purchase.

Through PostgreSQL, SQL, Excel, and Power BI, the project demonstrates an end-to-end data analytics workflow, from data preparation and validation to analytical reporting and business recommendations.

14. Project Deliverables and Resources

Project Repository: https://github.com/KumarAsishSahoo/E-COMMERCE-SALES-ANALYTICS.git

End of Report

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